

LabRack Vision QA

YOLO11s/960 + OpenCV · staged, non-patient images · human review required

Rack checks are repetitive—and still need judgment



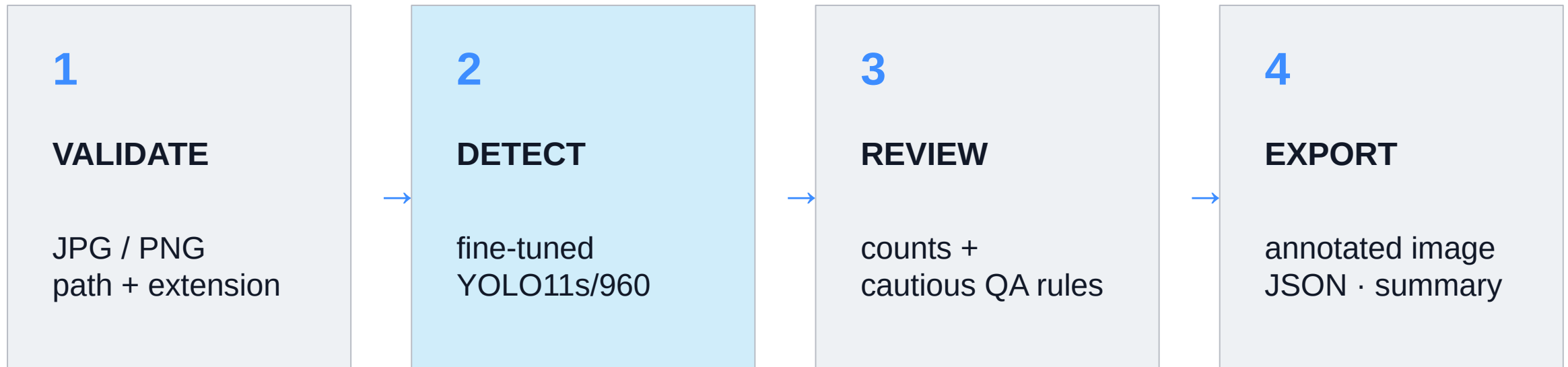
The prototype

Finds racks, caps, and possible empty slots in a still image.

The boundary

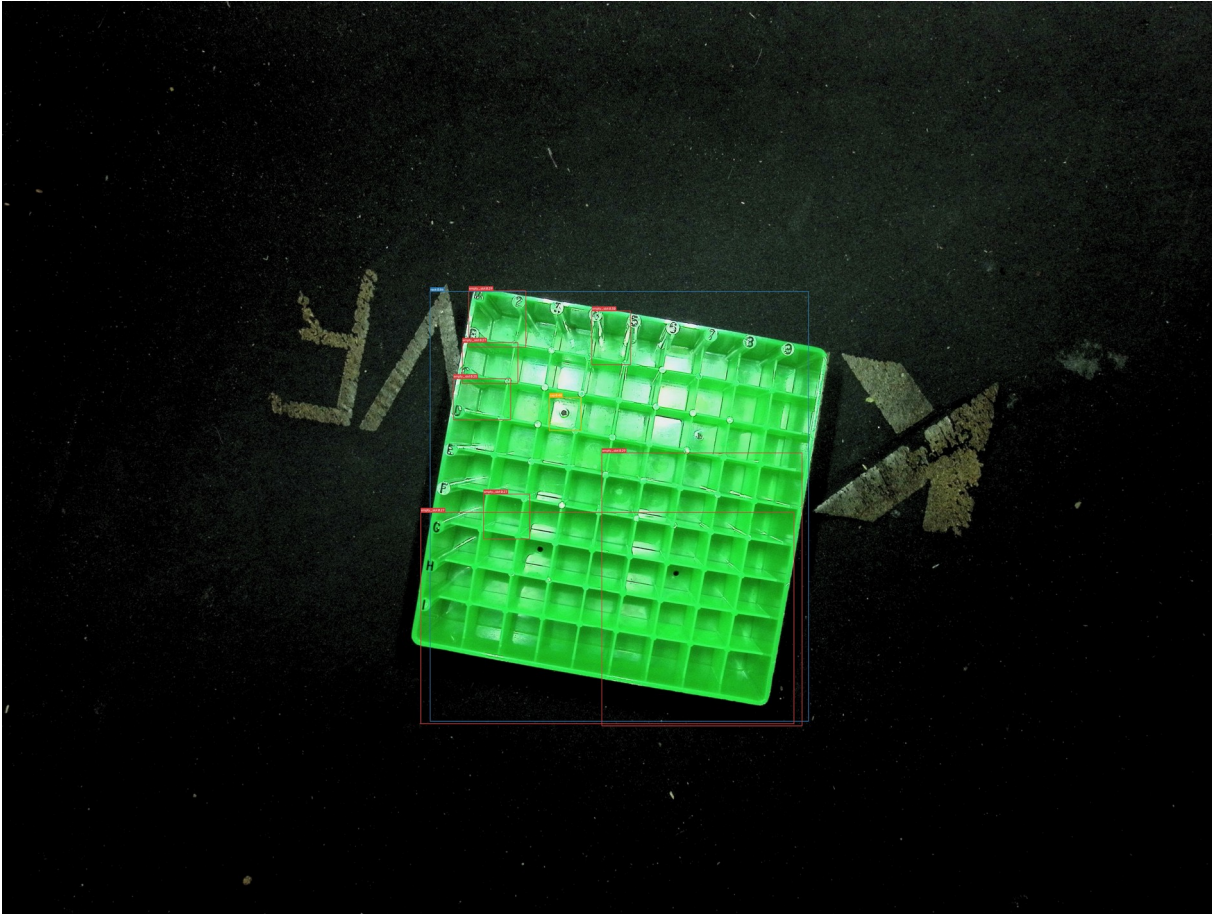
Every QA finding is a possible issue requiring human review. No patient data. No clinical claims.

What we built: one image becomes three artifacts



A fine-tuned detector feeds cautious QA rules and three reviewable outputs.

Demo: real input produces image, JSON, and review output



CLI SUMMARY

rack	1
tube	0
cap	1
empty_slot	7

8 detections below 0.50
→ **manual review recommended**

Detection step: 0.732 s

How it works: transfer-learned YOLO11s at 960 px

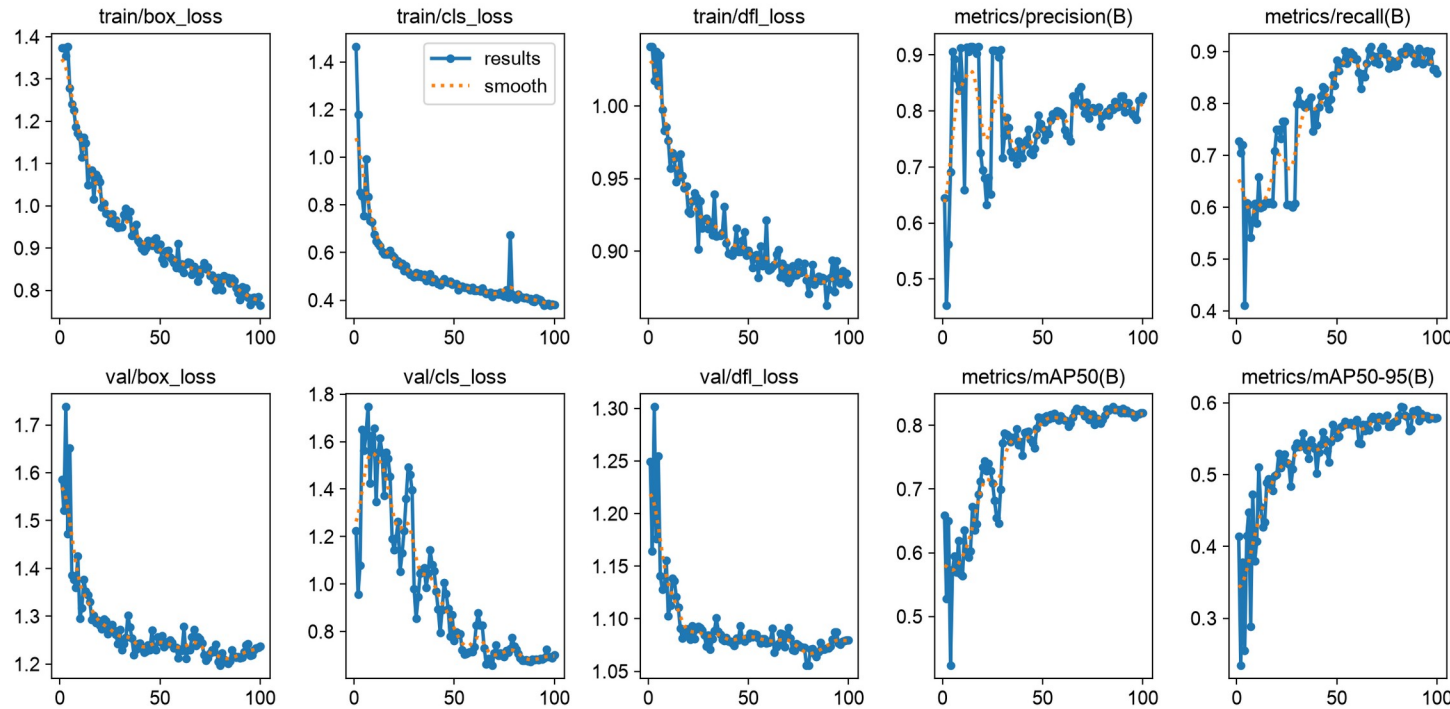
RUN

100 epochs
960 px · batch 2
YOLO11s

HARDWARE

Apple M3 Pro
PyTorch MPS

MPS emitted non-deterministic-
operation warnings.



A grouped split keeps adjacent rack images apart

140

staged images
partial Roboflow import

125

TRAIN · Rack_A/B

5

VALIDATE · Rack_C

10

TEST · Rack_D

Current annotation inventory

rack 133 cap 7,391 empty_slot 3,339

Kate Leemann · staged photos · annotations · testing methodology

Results versus our Blueprint targets

0.937

target ≥ 0.75
test mAP50

2.07 s

target < 3 s
full CLI wall time

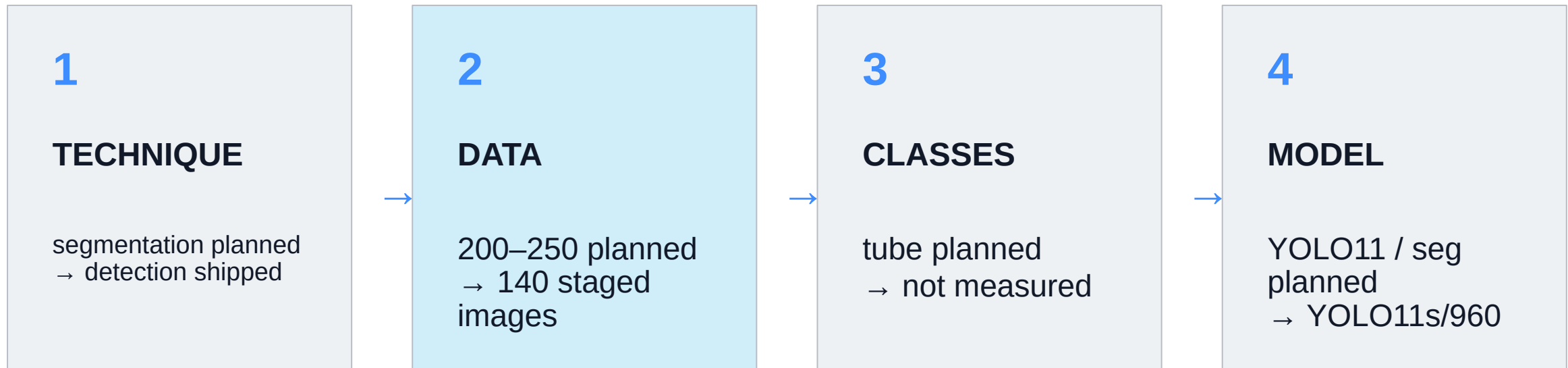
3 outputs

target: annotated
image + JSON + summary

Rack_D test: 10 staged images · 822 instances · mAP50–95 0.601

Both numeric targets were met; failure cases still require human review.

What changed from the Blueprint



The scope changed to fit the staged data; the end-to-end promise stayed intact.

Three fixes made the prototype work

960 px

small empty slots
needed more detail

GROUPED

rack-level split
reduced leakage

BATCH 2

MPS smoke tests
proved the model fit

Resolution, grouping, and memory checks were changed one at a time.

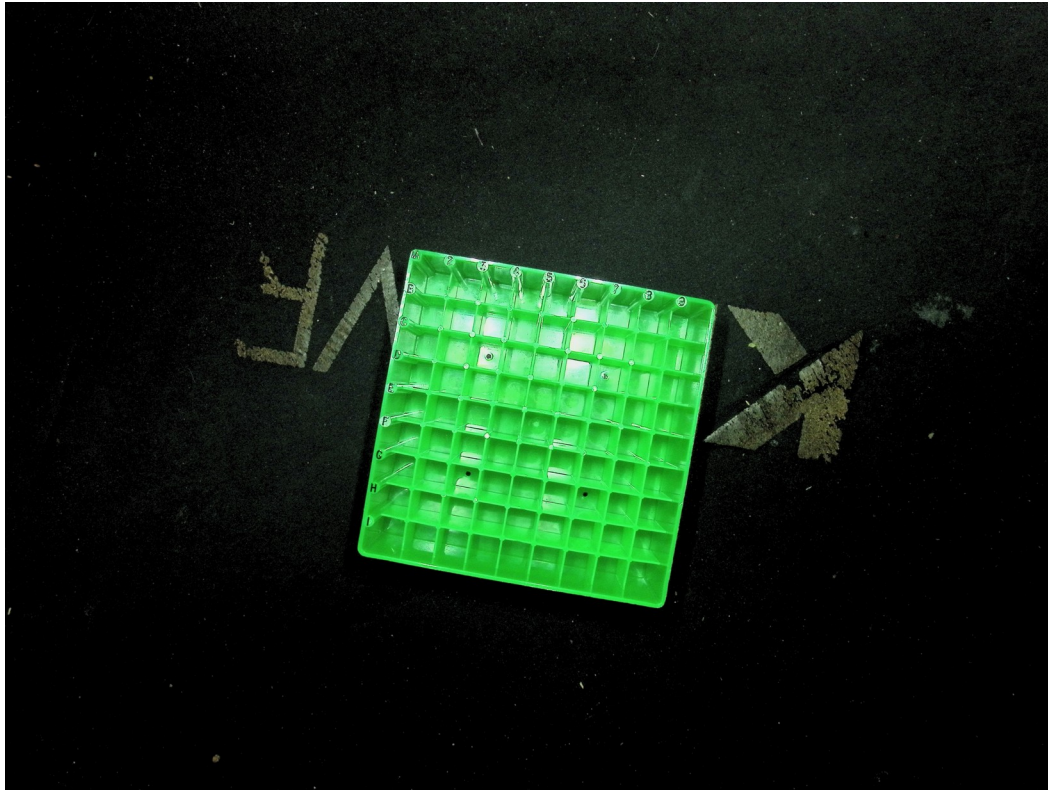
MPS reruns may vary slightly; every QA finding still requires human review.

The pipeline works; the dataset must grow.

- 1 grouped splits protect evaluation
- 2 higher resolution helps small slots
- 3 tube needs clearer staged images
- 4 next: new rack group + completed labels

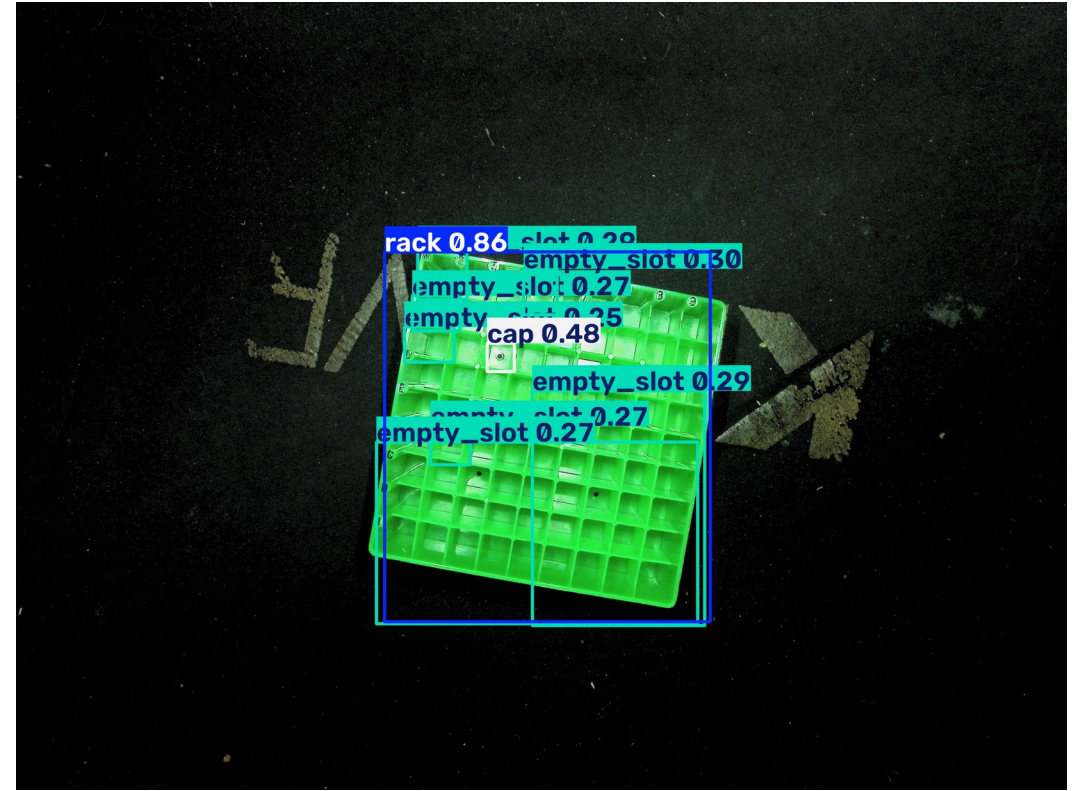
Human review required.
[github.com/RTaylor89/](https://github.com/RTaylor89/labrack-vision-qa)
[labrack-vision-qa](https://github.com/RTaylor89/labrack-vision-qa)

Failure case: 7 of 81 empty slots detected



KATE'S ANNOTATION

81 empty · 0 caps



MODEL OUTPUT

7 empty · 1 cap